

GNSS Compass

The GNSS Compass is a plug-and-play GNSS/INS navigation and heading solution.

Roll & Pitch	0.4 °
Position Accuracy (RTK)	0.01 m
Heading	0.2 °
Update Rate	200 Hz

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GNSS Compass is an all-in-one GNSS/INS navigation and heading solution. It provides accurate dual-antenna GPS-based heading that is not subject to magnetic interference and can maintain accurate heading during GNSS outages of up to 20 minutes. It features high accuracy 1cm RTK positioning and is plug and play with NMEA 0183, NMEA 2000, and Ethernet interfaces.

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Features

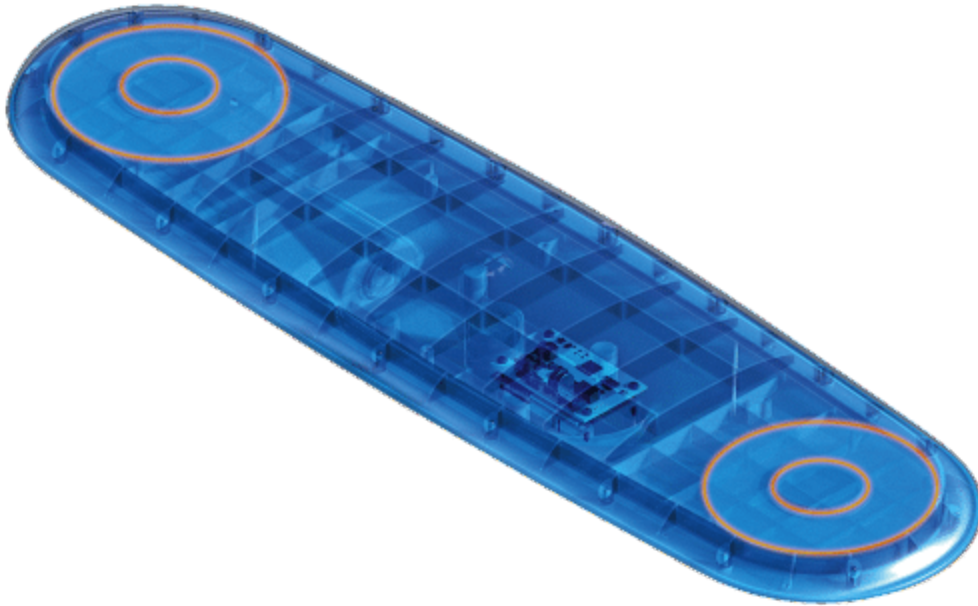
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Features



Plug-And-Play

GNSS Compass is a fully plug-and-play solution with support for both NMEA 2000 and NMEA 0183. There is no configuration, setup or calibration required. Reliable position and heading in minutes.



GNSS Heading

GNSS Compass uses dual GNSS antennas to reliably determine true heading with high accuracy. It doesn't require motion to work and it

isn't affected by magnetic interference.



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Interface Options

GNSS Compass is available in both a serial variant and a power over ethernet variant. The serial variant supports both NMEA 2000 and NMEA 0183 for plug-and-play connection to existing marine systems, while the power over ethernet variant allows simplified power and network cabling with maximum flexibility for new applications.



Dual Frequency RTK GNSS

GNSS Compass features L1/L2 RTK to deliver real time position accuracy of 10 mm. It supports all of the current and future satellite navigation systems including GPS, GLONASS, GALILEO and BeiDou.



PTP-NTP

Time Server

GNSS Compass features a GPS disciplined oscillator allowing it to act as a high accuracy time reference for other systems. The power over

ethernet variant supports both PTP and NTP for precise time
synchronization across our whole network.

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Version 3.4.0 | 10 Jun 2026

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Reference Manual

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Ethernet 3D Model

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INS Manager – Desktop is used for testing, configuration, firmware management and logging with Advanced Navigation's INS devices. Java 11 is required to run *INS Manager – Desktop*. [Adoptium JRE 11](#) is the recommended Java installer. Refer to respective INS Manuals for the user guide.

GNSS Compass Firmware

Version 2.47 | 5 Apr 2024



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The firmware for all GNSS Compass hardware versions. This can be loaded onto GNSS Compass using the web interface or the GNSS Compass Manager software. The change log can be found in the GNSS Compass reference manual.

GNSS Compass SDK

Version 7.3 | 28 Aug 2024



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The GNSS Compass SDK provides example source code for interfacing with Certus through the AN Packet Protocol. The languages provided are C/C++, Java, and .Net C#.

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